ABSTRACT OF THE DISCLOSURE

A system and method for remotely imaging the interior of a target space, such as a building or vehicle, using a pair of coordinated mobile units. The system comprises a first mobile unit equipped with a signal emitter (transmitter) and a second mobile unit equipped with a synchronized signal sensor (receiver). The two units are positioned on opposing sides of a target space. The transmitter emits a non-ionizing radiation signal (e.g., millimeter-wave) which penetrates the walls or fuselage of the target space. The receiver captures the attenuated signal that has passed through the interior of the space. By analyzing the differential attenuation of the signal, a two-dimensional or three-dimensional image of the contents within the target space, including metallic objects or human occupants, can be computationally reconstructed. The system is designed for rapid, non-invasive inspection of structures from a standoff distance.